```
clc;
clear;
close all;
```

FIR FILTER PARAMETERS

```
b = [0.0625 0.2500 0.3750 0.2500 0.0625];
Fs = 1000;
N = 128;

x = [1 zeros(1, N-1)];
n = 0:N-1;
```

1. DIRECT REALIZATION

```
yd = filter(b, 1, x);
```

2. CASCADE REALIZATION

```
z = roots(b);
g = b(1);
yc = g * x;

for k = 1:length(z)
    yc = filter([1 -z(k)], 1, yc);
end

yc = real(yc);
```

3. SYMMETRY AND LINEAR PHASE

```
symErr = max(abs(b - fliplr(b)));
D = (length(b) - 1) / 2;

if symErr < 1e-12
    symmetry = 'Symmetric';
else
    symmetry = 'Not Symmetric';
end
```

4. FREQUENCY, PHASE AND GROUP DELAY

```
[H, w] = freqz(b, 1, 1024);
phase = unwrap(angle(H));
gd = grpdelay(b, 1, 1024);
```

5. NUMERICAL COMPARISON

```
err = max(abs(yd - yc));
peakOutput = max(abs(yd));
```

```
if peakOutput > 0
    dev = (err / peakOutput) * 100;
else
    dev = 0;
end
```

6. DISPLAY RESULTS

```
fprintf('\n=============================================\n');
fprintf('       FIR FILTER REALIZATION RESULTS\n');
fprintf('=============================================\n');

fprintf('\nFIR Coefficients:\n');
disp(b);

fprintf('Sampling Frequency = %d Hz\n', Fs);
fprintf('Number of Samples = %d\n', N);
fprintf('Number of Taps = %d\n', length(b));
fprintf('Coefficient Symmetry = %s\n', symmetry);
fprintf('Symmetry Error = %.3e\n', symErr);
fprintf('Group Delay = %.2f samples\n', D);

fprintf('\nMaximum Error = %.3e\n', err);
fprintf('Percentage Deviation = %.3e %%\n', dev);
```

```
=============================================
       FIR FILTER REALIZATION RESULTS
=============================================

FIR Coefficients:
    0.0625    0.2500    0.3750    0.2500    0.0625

Sampling Frequency = 1000 Hz
Number of Samples = 128
Number of Taps = 5
Coefficient Symmetry = Symmetric
Symmetry Error = 0.000e+00
Group Delay = 2.00 samples

Maximum Error = 8.327e-16
Percentage Deviation = 2.220e-13 %
```

7. FIRST 10 IMPULSE RESPONSE VALUES

```
fprintf('\n=============================================\n');
fprintf('      FIRST 10 IMPULSE RESPONSE VALUES\n');
fprintf('=============================================\n');

fprintf('%4s %14s %14s\n', 'n', 'Direct', 'Cascade');

for k = 1:min(10, N)
```

```
    fprintf('%4d %14.8f %14.8f\n', ...
        n(k), yd(k), yc(k));
end
```

```
============================================
      FIRST 10 IMPULSE RESPONSE VALUES
============================================
   n         Direct        Cascade
   0     0.06250000     0.06250000
   1     0.25000000     0.25000000
   2     0.37500000     0.37500000
   3     0.25000000     0.25000000
   4     0.06250000     0.06250000
   5     0.00000000     0.00000000
   6     0.00000000     0.00000000
   7     0.00000000     0.00000000
   8     0.00000000     0.00000000
   9     0.00000000     0.00000000
```

8. CASCADE REALIZATION ZEROS

```
fprintf('\n============================================\n');
fprintf('         CASCADE REALIZATION ZEROS\n');
fprintf('============================================\n');

disp(z);
```

```
============================================
         CASCADE REALIZATION ZEROS
============================================
  -1.0002 + 0.0000i
  -1.0000 + 0.0002i
  -1.0000 - 0.0002i
  -0.9998 + 0.0000i
```

9. OUTPUT PARAMETER TABLE

```
Parameter = {
    'Direct realization output'
    'Cascade realization output'
    'Impulse response'
    'Phase response'
    'Comparison of realizations'
};

Observation = {
    'Direct FIR filtering'
    'Cascade of first-order sections'
    'Equal to FIR coefficient sequence'
    'Linear phase'
```

```
    'Equivalent transfer-function behavior'
};

Value = {
    sprintf('[%.4f %.4f %.4f %.4f %.4f]', yd(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]', yc(1:5))
    sprintf('[%.4f %.4f %.4f %.4f %.4f]', yd(1:5))
    sprintf('Group delay = %.2f samples', D)
    sprintf('Max error = %.3e; Deviation = %.3e %%', err, dev)
};

Unit_Note = {
    'Amplitude'
    'Amplitude'
    'h[n]'
    'Samples'
    'Numerical precision'
};

Output_Table = table(Parameter, Observation, Value, Unit_Note);

fprintf('\n===========================================\n');
fprintf('           OUTPUT PARAMETER TABLE\n');
fprintf('===========================================\n');

disp(Output_Table);
```

```
===========================================
           OUTPUT PARAMETER TABLE
===========================================
              Parameter
Observation
Value                                  Unit_Note
    ___________________________________
_____________________________________________
_____________________________________________     ________________________

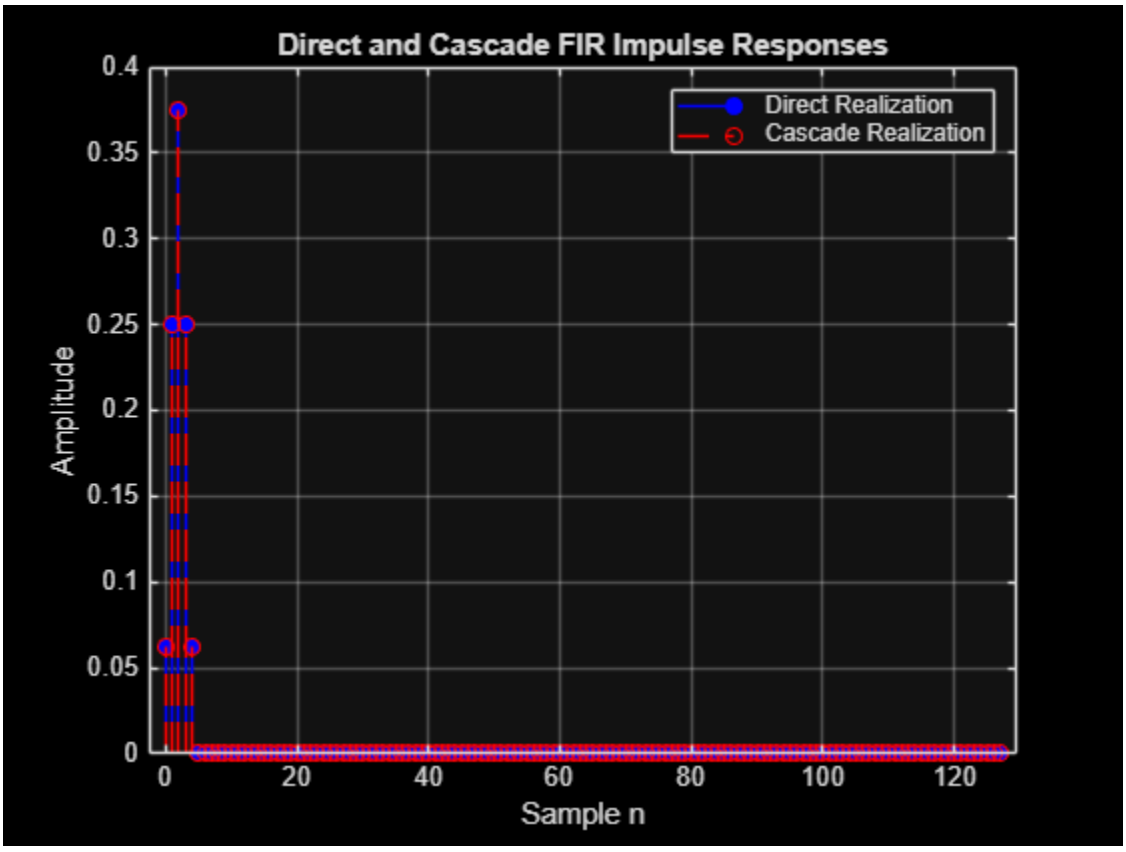
    {'Direct realization output' }    {'Direct FIR
filtering'                  }    {'[0.0625 0.2500 0.3750 0.2500
0.0625]'          }    {'Amplitude'          }
    {'Cascade realization output'}    {'Cascade of first-order
sections'      }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'          }
{'Amplitude'          }
    {'Impulse response'          }    {'Equal to FIR coefficient
sequence'    }    {'[0.0625 0.2500 0.3750 0.2500 0.0625]'          }
{'h[n]'               }
    {'Phase response'            }    {'Linear
phase'                         }    {'Group delay = 2.00
samples'                    }    {'Samples'            }
    {'Comparison of realizations'}    {'Equivalent transfer-function
behavior'}    {'Max error = 8.327e-16; Deviation = 2.220e-13 %'}
{'Numerical precision'}
```

10. GRAPH 1: DIRECT VS CASCADE IMPULSE RESPONSE

```
figure('Name', 'Impulse Response Comparison');

stem(n, yd, 'b', 'filled');
hold on;
stem(n, yc, 'r--');

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Direct and Cascade FIR Impulse Responses');
legend('Direct Realization', 'Cascade Realization');
```



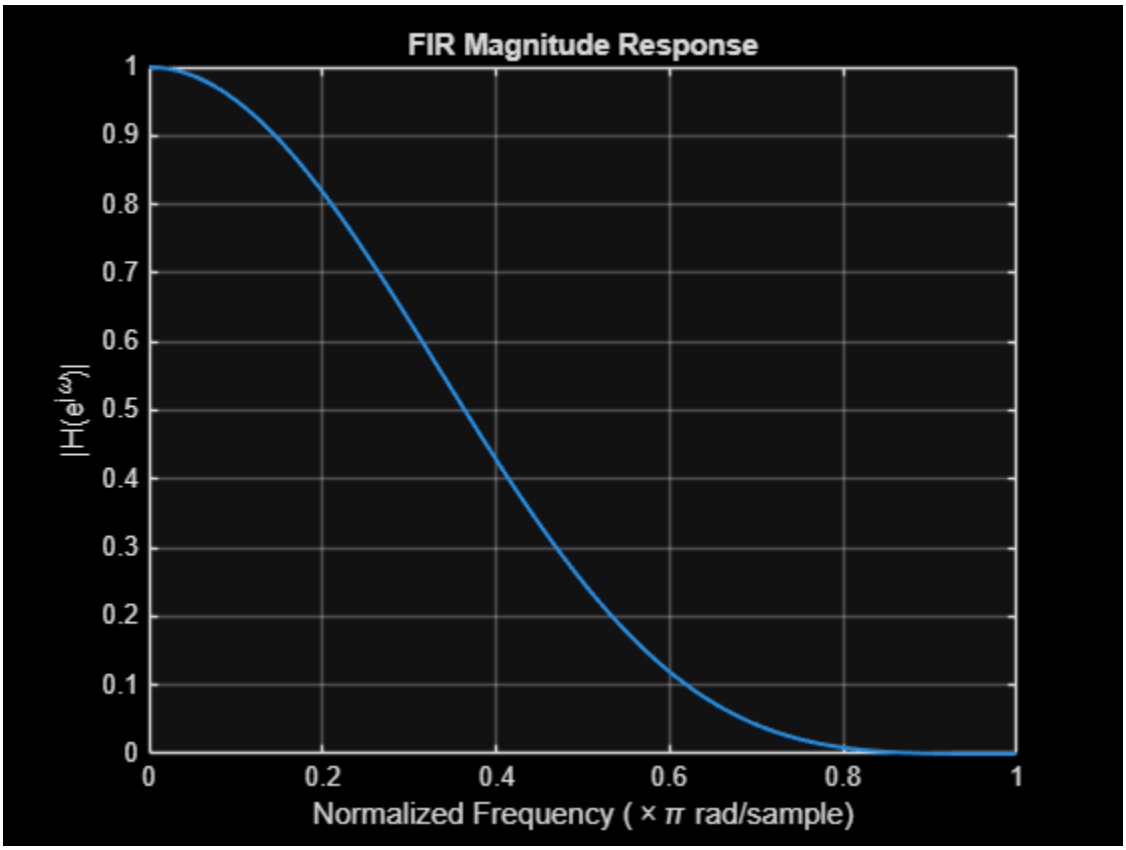
11. GRAPH 2: MAGNITUDE RESPONSE

```
figure('Name', 'FIR Magnitude Response');

plot(w/pi, abs(H), 'LineWidth', 1.5);

grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('|H(e^{j\omega})|');
```

```
title('FIR Magnitude Response');
xlim([0 1]);
```

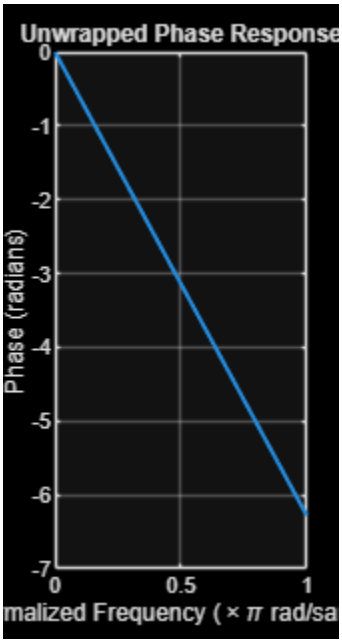


12. GRAPH 3: PHASE RESPONSE

```
figure('Name', 'FIR Phase Response');

plot(w/pi, phase, 'LineWidth', 1.5);

grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Phase (radians)');
title('Unwrapped Phase Response');
xlim([0 1]);
```

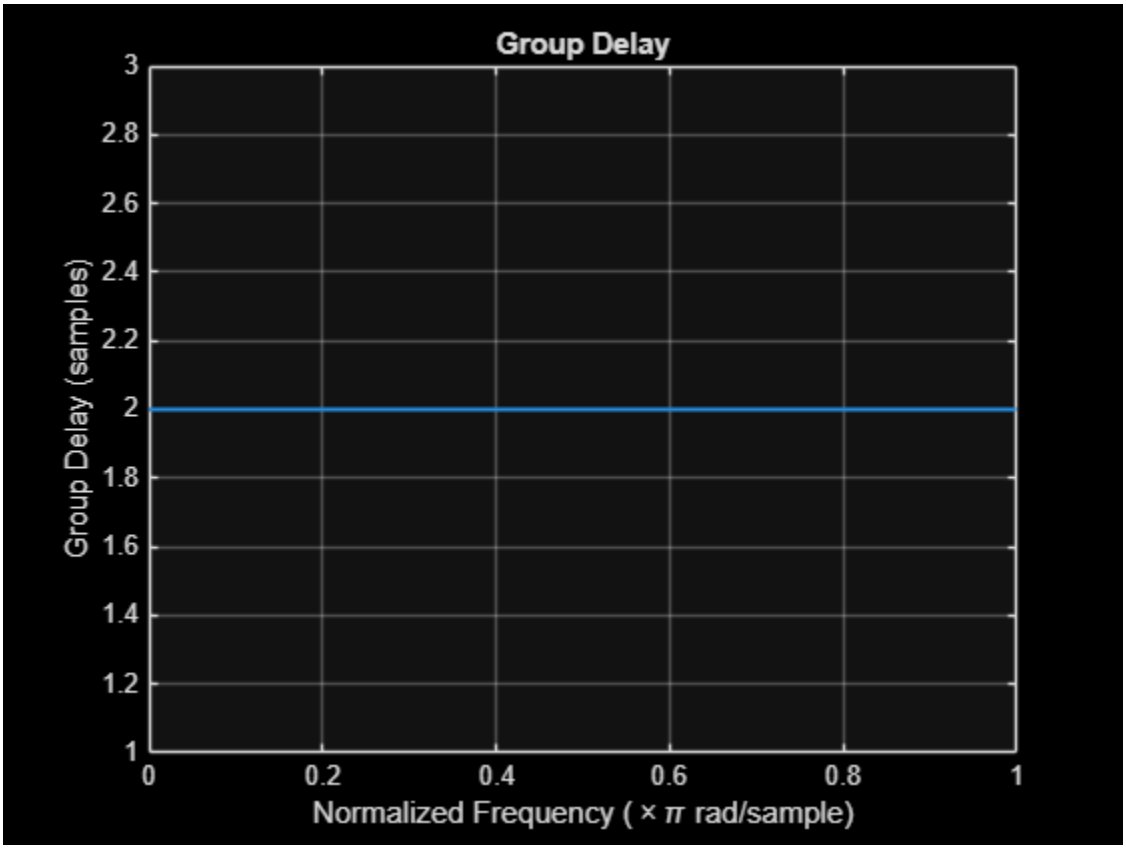


13. GRAPH 4: GROUP DELAY

```
figure('Name', 'FIR Group Delay');

plot(w/pi, gd, 'LineWidth', 1.5);

grid on;
xlabel('Normalized Frequency (\times\pi rad/sample)');
ylabel('Group Delay (samples)');
title('Group Delay');
xlim([0 1]);
```

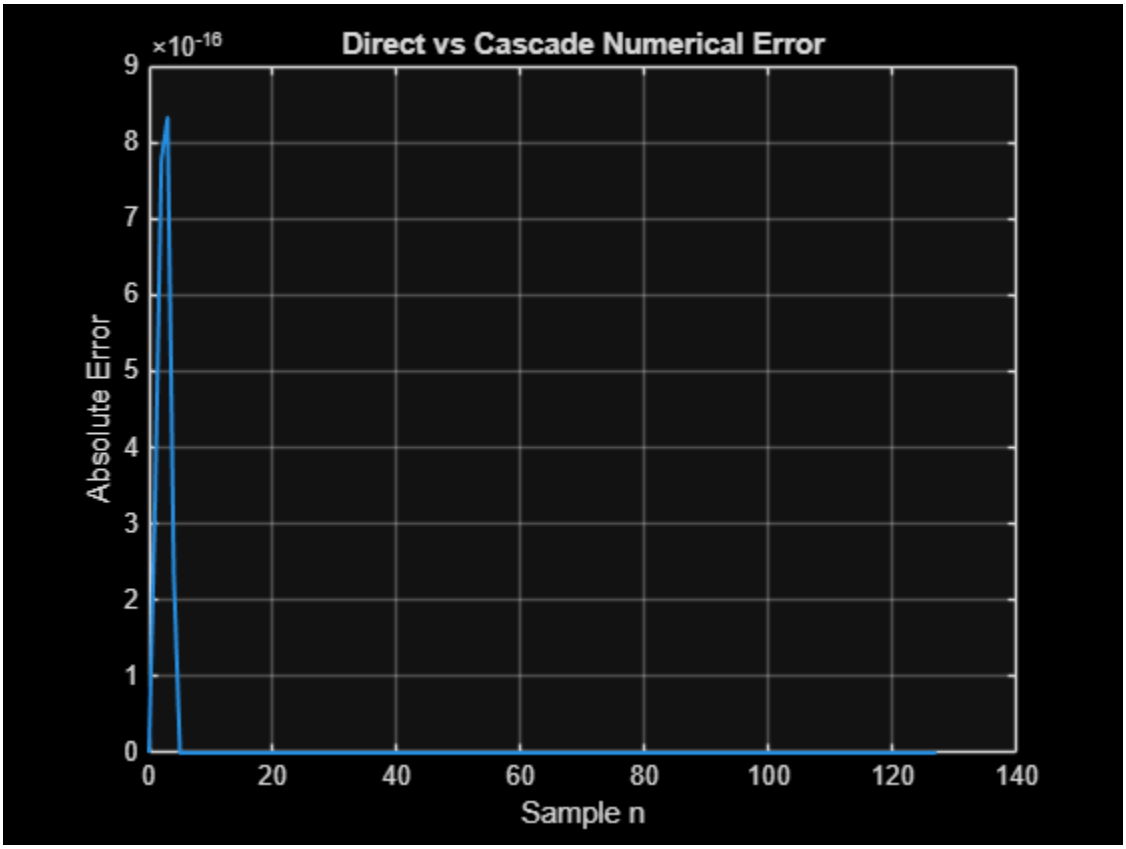


14. GRAPH 5: NUMERICAL ERROR

```
figure('Name', 'Numerical Error');

plot(n, abs(yd - yc), 'LineWidth', 1.5);

grid on;
xlabel('Sample n');
ylabel('Absolute Error');
title('Direct vs Cascade Numerical Error');
```



15. FINAL RESULT

```
fprintf('\n=============================================\n');
fprintf('                 FINAL RESULT\n');
fprintf('=============================================\n');

fprintf('1. FIR filter realized using Direct structure.\n');
fprintf('2. FIR filter realized using Cascade structure.\n');
fprintf('3. Coefficients are %s.\n', symmetry);
fprintf('4. Linear-phase group delay = %.2f samples.\n', D);
fprintf('5. Maximum realization error = %.3e.\n', err);
fprintf('6. Percentage deviation = %.3e %%.\n', dev);
fprintf('7. Direct and Cascade responses are equivalent.\n');
```

```
=============================================
                 FINAL RESULT
=============================================
1. FIR filter realized using Direct structure.
2. FIR filter realized using Cascade structure.
3. Coefficients are Symmetric.
4. Linear-phase group delay = 2.00 samples.
5. Maximum realization error = 8.327e-16.
6. Percentage deviation = 2.220e-13 %.
7. Direct and Cascade responses are equivalent.
```

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